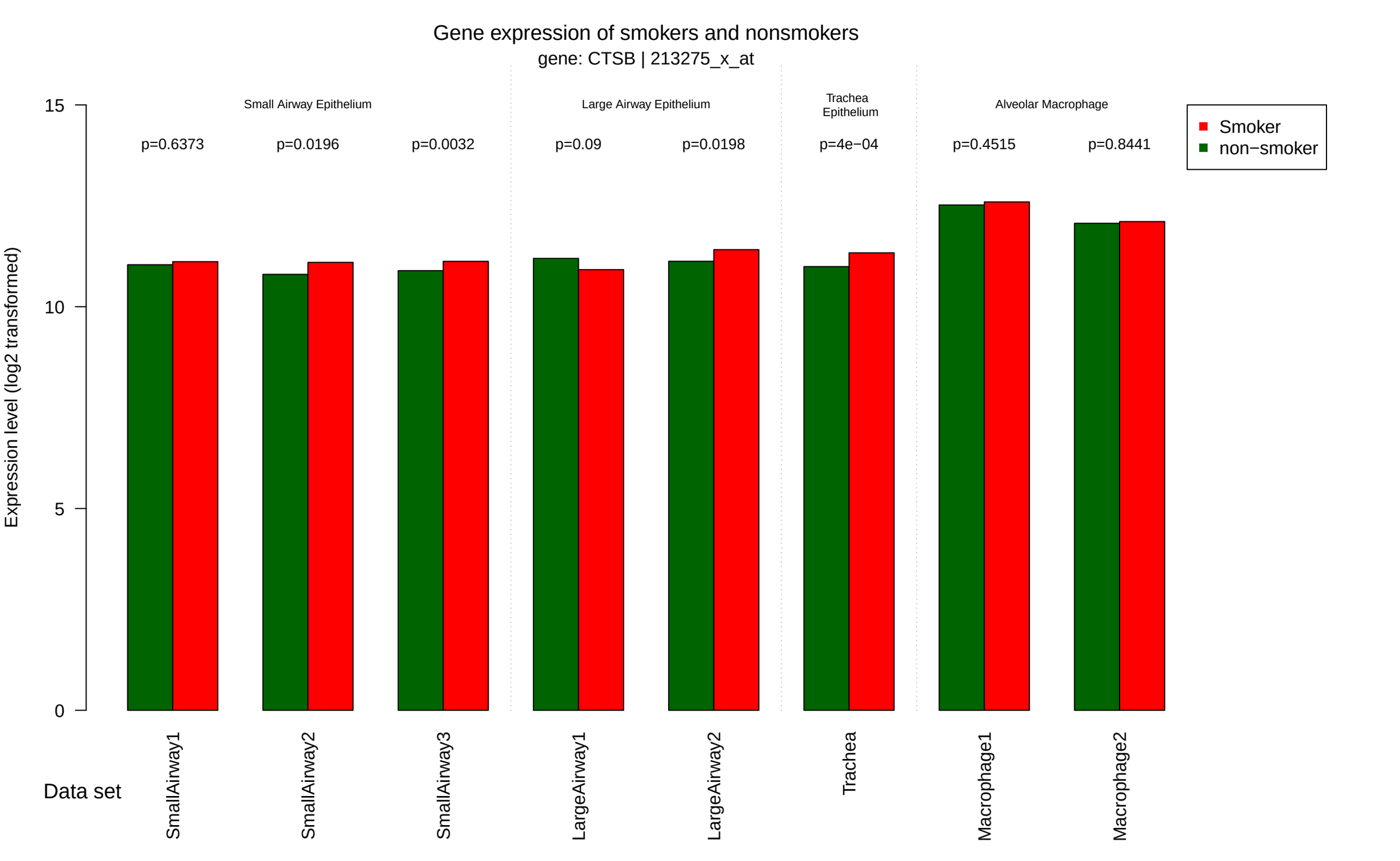
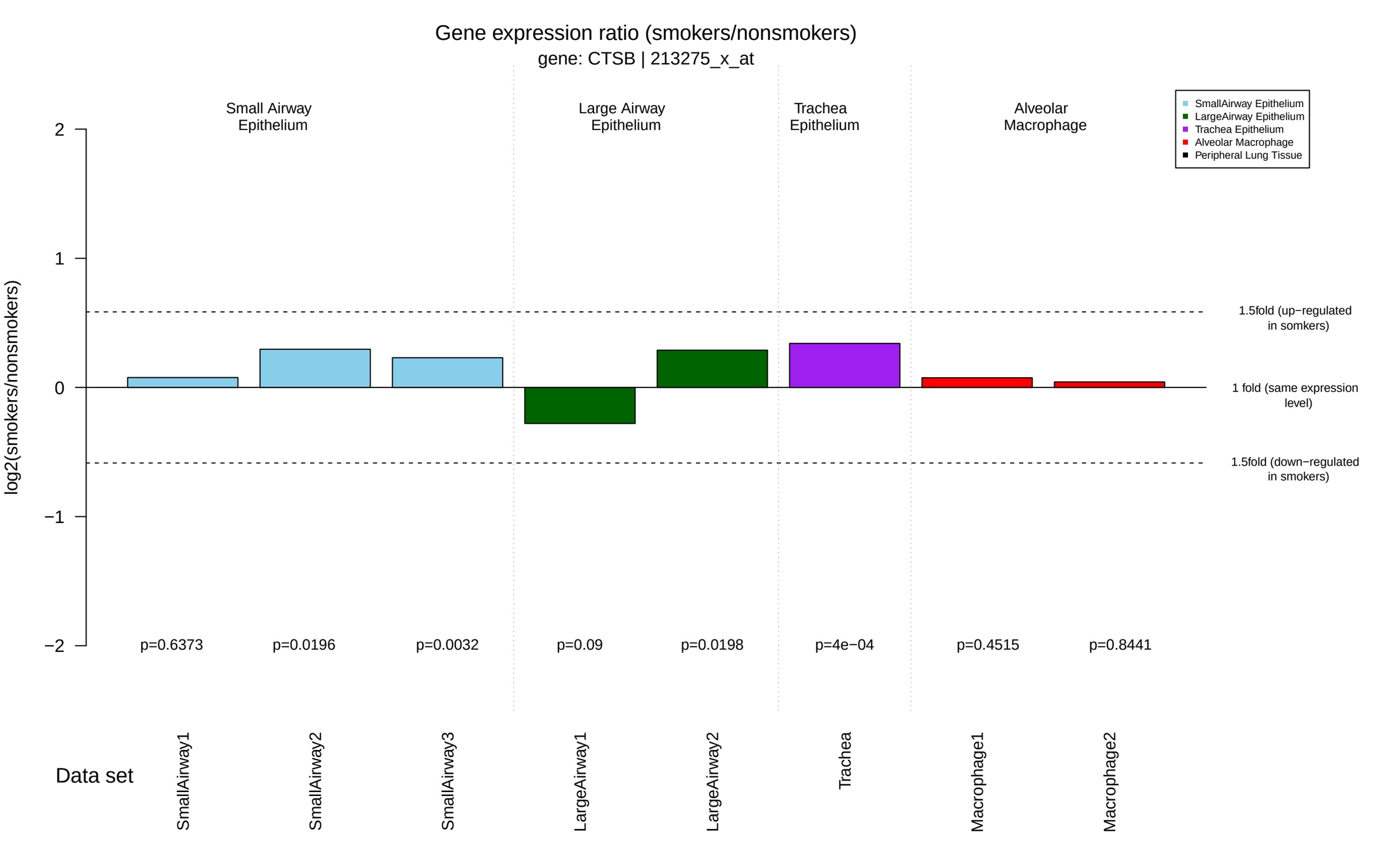
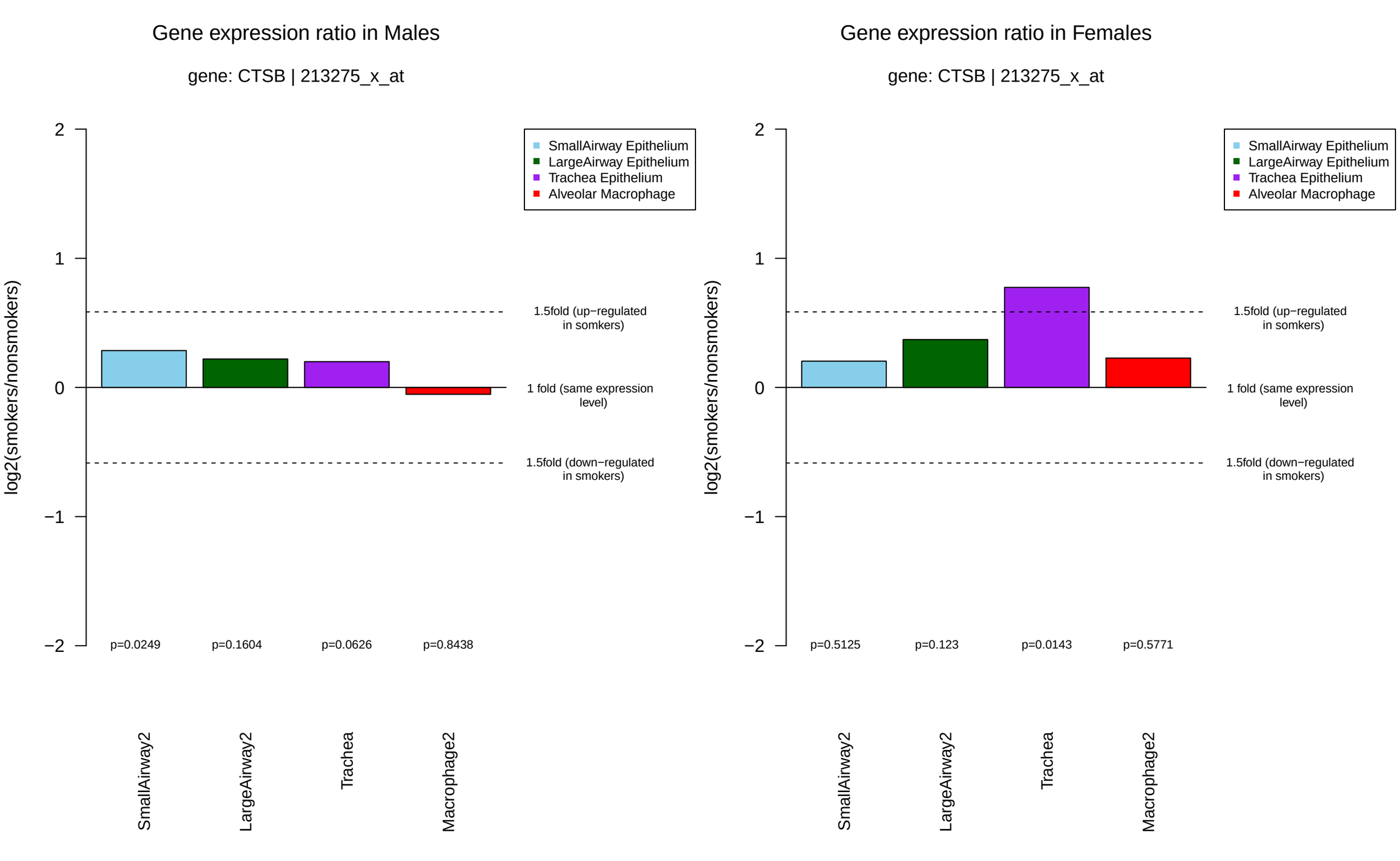
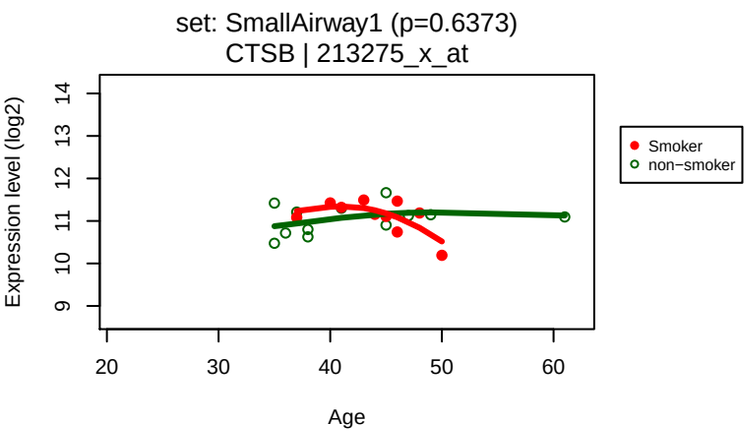
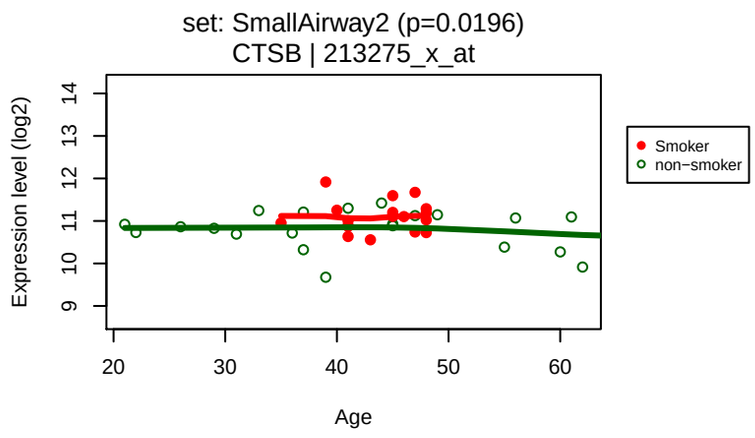
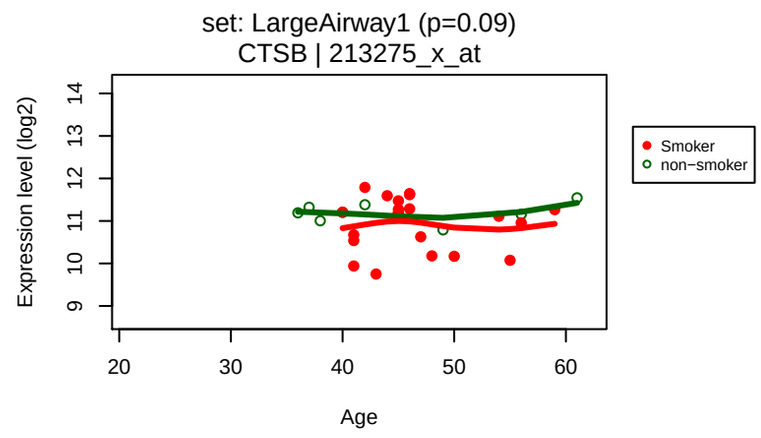

Gene expression of smokers and nonsmokers
gene: CTSB | 213275_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6373
p=0.0196
p=0.0032
p=0.09
p=0.0198
p=4e−04
p=0.4515
p=0.8441
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

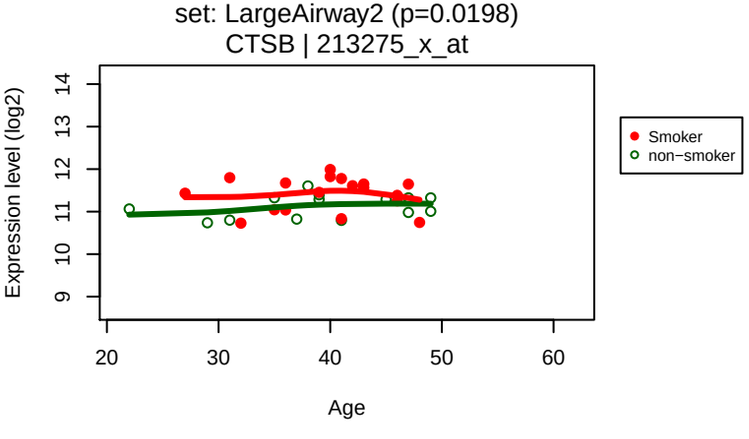

Gene expression ratio (smokers/nonsmokers)
gene: CTSB | 213275_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6373
p=0.0196
p=0.0032
p=0.09
p=0.0198
p=4e−04
p=0.4515
p=0.8441
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

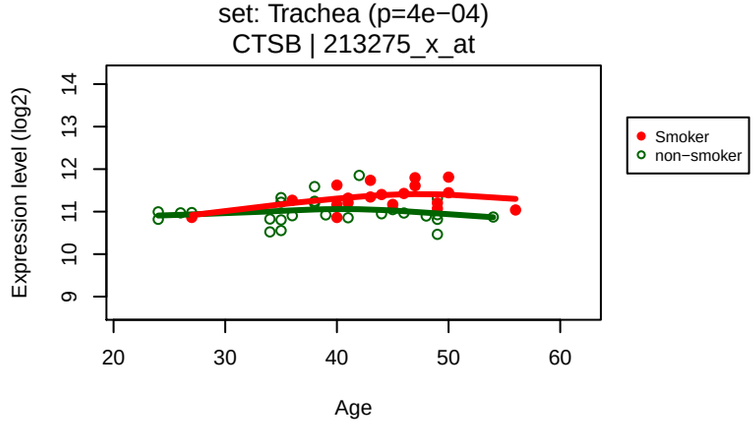

Gene expression ratio in Males
gene: CTSB | 213275_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0249
p=0.1604
p=0.0626
p=0.8438
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CTSB | 213275_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5125
p=0.123
p=0.0143
p=0.5771
SmallAirway2
LargeAirway2
Trachea
Macrophage2

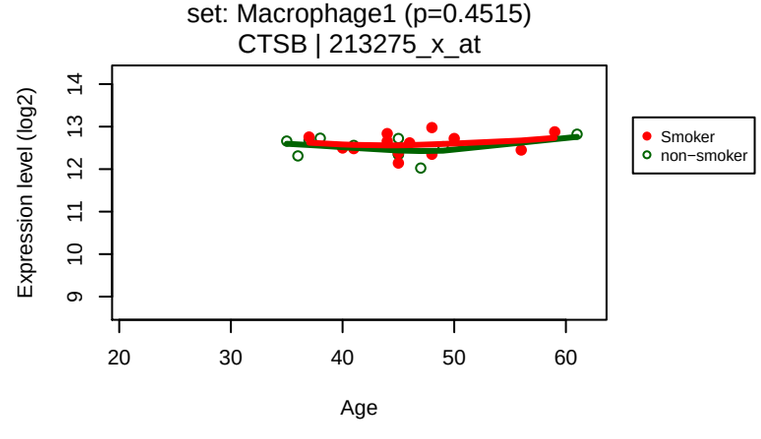

set: SmallAirway1 (p=0.6373)
CTSB | 213275_x_at
Expression level (log2)
Age
Smoker
non−smoker

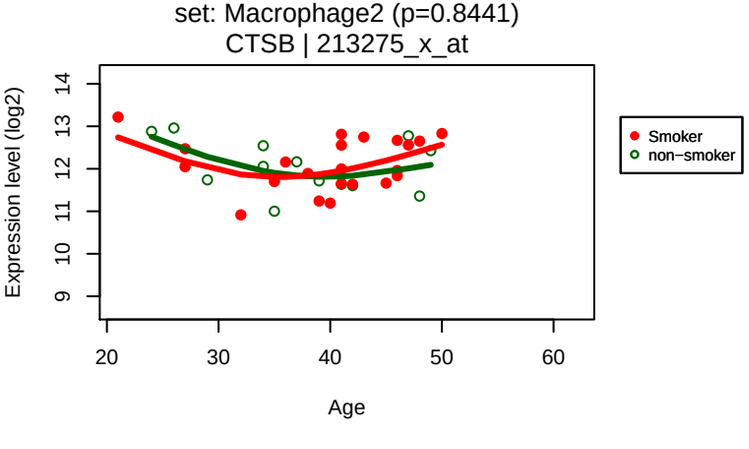

set: SmallAirway2 (p=0.0196)
CTSB | 213275_x_at
Expression level (log2)
Age
Smoker
non−smoker

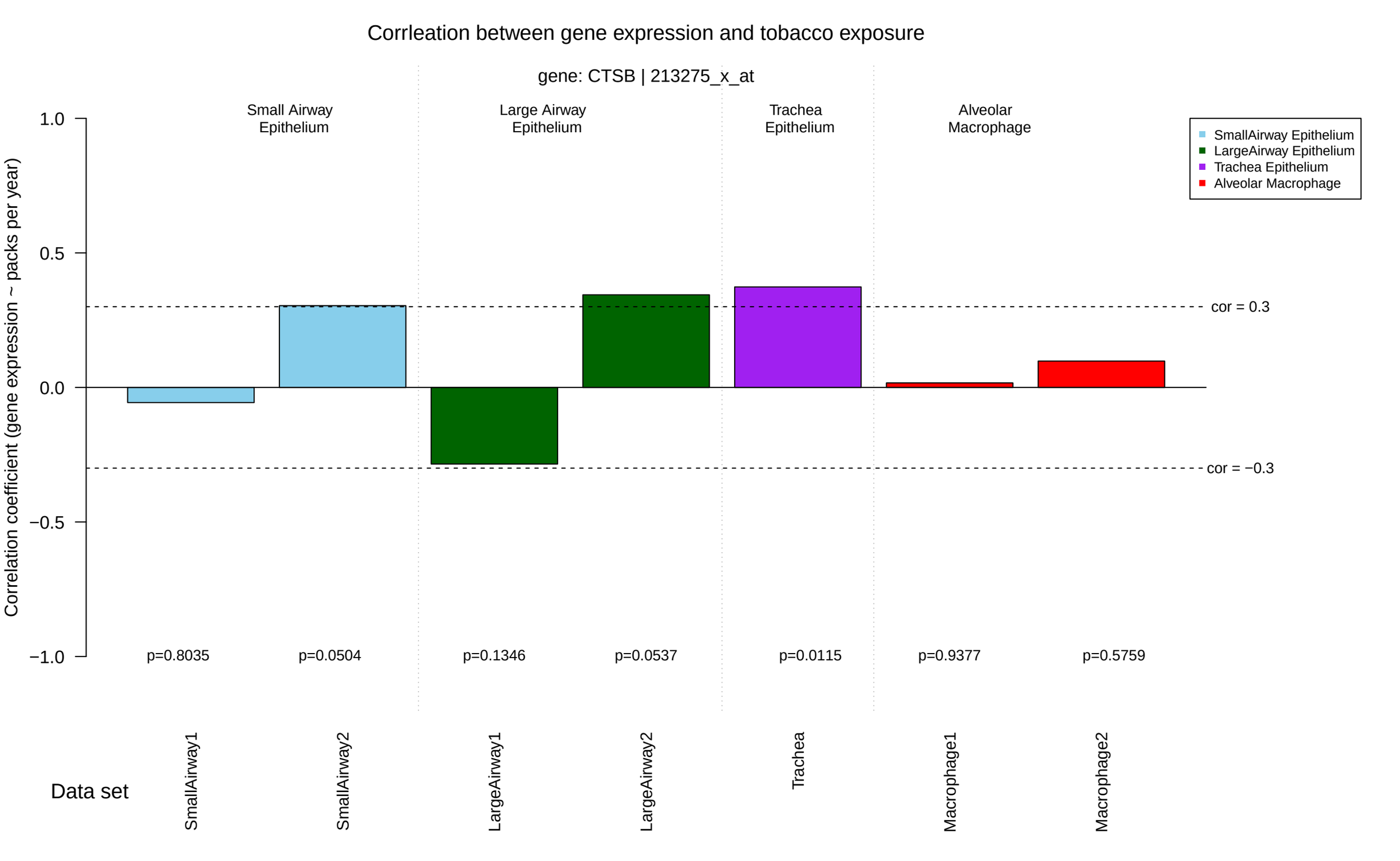

set: LargeAirway1 (p=0.09)
CTSB | 213275_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0198)
CTSB | 213275_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=4e−04)
CTSB | 213275_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4515)
CTSB | 213275_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8441)
CTSB | 213275_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CTSB | 213275_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8035
p=0.0504
p=0.1346
p=0.0537
p=0.0115
p=0.9377
p=0.5759
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2